CHAPTER 19

NATURAL VEGETATION

CHAPTER ESSENCE AT A GLANCE:
Examine the five major forest types of India (Tropical Evergreen, Tropical Deciduous, Tropical Thorn, Montane, Littoral and Mangrove), National Forest Policy benchmarks, social forestry, and wildlife preservation networks.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Tropical Evergreen and Semi-Evergreen Formations:** Flourish in areas with annual rainfall > 200 cm and mean annual temperature > 22 degrees Celsius (Western Ghats, North-East); dense multi-layered canopy exceeding 60 m height; valuable timbers include rosewood, mahogany, and ebony.

• **Tropical Deciduous (Monsoon) Forests:** The most widespread forest formation in India (rainfall 70 to 200 cm); shed leaves for 6 to 8 weeks in dry spring; divided into Moist Deciduous (teak, sal, shisham, sandalwood) and Dry Deciduous (tendu, palas, amaltas, khair).

• **Thorn and Montane Altitudinal Succession:** Thorn forests occupy semi-arid regions (< 50 cm rainfall, acacias, euphorbias); Montane forests display complete altitudinal zonation from wet temperate broadleaved oaks to coniferous pines and deodars, to alpine pastures.

• **Littoral Mangrove Forests and Conservation:** Halophytic tidal forests flourishing in estuarine deltas (Ganga-Brahmaputra Sundarbans); adapted with breathing pneumatophores and stilt roots; protected under National Forest Policy (targeting 33 percent forest cover) and Biosphere Reserve networks.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Forest Rainfall Thresholds: Evergreen } (>200\text{ cm}) \longleftrightarrow \text{Moist } (100 \text{ to } 200) \longleftrightarrow \text{Dry } (70 \text{ to } 100\text{ cm})$$

Precipitation-Bound Forest Typology Classification

$$\text{National Forest Policy Target} = \frac{\text{Forest Area}}{\text{Total Geographic Area}} \geq 33.3\% \quad (\text{Hilly Regions: } 67\%)$$

Statutory Ecological Forest Cover Coverage Targets

$$\text{Montane Succession: Subtropical } (1000 \text{ to } 2000\text{ m}) \rightarrow \text{Temperate } (2000 \text{ to } 3000\text{ m}) \rightarrow \text{Alpine } (>3000\text{ m})$$

Altitudinal Climatological Zonation of Himalayan Flora

$$\text{Mangrove Production} = f(\text{Tidal Inundation, Salinity Gradient, Estuarine Silt Influx})$$

Estuarine Environmental Function Governing Mangrove Biomass

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Tropical Deciduous forests are the most widespread forest type in India, not Tropical Evergreen forests; deciduous forests cover nearly two-thirds of India's total forested area.

• **Alert:** Mangrove trees possess specialized aerial respiratory roots called pneumatophores that project vertically upwards out of the saline water and hypoxic mud to take in atmospheric oxygen.

• **Alert:** Social Forestry is subdivided into three distinct categories: Urban forestry, Rural forestry (which includes agro-forestry and community forestry), and Farm forestry, designed to reduce pressure on natural reserve forests.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Name the most widespread forest type in India and state its typical annual rainfall range. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **Most Widespread Forest:** Tropical Deciduous Forest (popularly known as Monsoon Forest). **[1 Mark]**

• **Rainfall Range:** Flourishes in regions receiving annual rainfall between 70 cm and 200 cm. **[1 Mark]**

2. What are Pneumatophores? In which specific vegetation type are they found? **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **Definition:** Specialized erect aerial breathing roots that grow vertically upwards above the surface of tidal water and waterlogged mud to obtain oxygen for root respiration. **[1 Mark]**

• **Vegetation Type:** Found in Littoral and Swamp forests (Mangrove vegetation), such as in the Sundarbans. **[1 Mark]**

3. Distinguish between Tropical Evergreen Forests and Tropical Deciduous Forests on the basis of rainfall, canopy, and tree species. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Rainfall Requirement:** Evergreen forests require high annual rainfall exceeding 200 cm; Deciduous forests grow in regions receiving 70 to 200 cm rainfall. **[1 Mark]**

• **Canopy and Leaf Shedding:** Evergreen forests have dense, multi-tiered canopies that never shed leaves simultaneously; Deciduous forests shed their leaves for 6 to 8 weeks in dry spring. **[1 Mark]**

• **Dominant Tree Species:** Evergreen timber includes rosewood, mahogany, ebony; Deciduous timber includes economically valuable teak, sal, shisham, and sandalwood. **[1 Mark]**

4. Explain the vertical zonation of natural vegetation observed in the Himalayan mountain ranges with increasing altitude. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **1000 to 2000 Metres:** Wet temperate forests characterized by broadleaved evergreen trees like evergreen oaks and chestnuts. **[1 Mark]**

• **1500 to 3000 Metres:** Temperate coniferous forests dominated by pine, deodar (valuable rot-resistant construction wood), spruce, and silver fir. **[1 Mark]**

• **Above 3000 Metres:** Transitions through silver fir and junipers into Alpine meadows and pastures (grazed by pastoral tribes like Gujjars and Bakarwals) before mosses and lichens at snow line. **[1 Mark]**

5. What is Social Forestry? Describe its three main components (urban, rural, and farm forestry). **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **Concept:** Management and protection of forests and afforestation on barren lands with the purpose of helping in the environmental, social, and rural development. **[1 Mark]**

• **Urban Forestry:** Raising and managing of trees on public and privately owned lands in and around urban centers such as parks, green belts, and roadside avenues. **[1 Mark]**

• **Rural Forestry:** Divided into Agro-forestry (growing trees and agricultural crops on the same land) and Community forestry (planting trees on public lands like village common pastures, temple grounds, and canals). **[1 Mark]**

• **Farm Forestry:** Farmers are encouraged to plant trees for commercial and non-commercial purposes on their own farmland margins and bunds. **[1 Mark]**

6. Discuss the principal objectives of the National Forest Policy (1988) of India and assess measures for wildlife conservation. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **Forest Cover Target:** Bringing 33 percent of total geographical land area of the nation under forest or tree cover, with a higher target of 67 percent in mountainous and hilly regions. **[1 Mark]**

• **Environmental Balance:** Maintenance of environmental stability and restoration of ecological balance adversely disturbed by deforestation. **[1 Mark]**

• **Soil and Water Conservation:** Checking soil erosion in catchments of rivers, lakes, and reservoirs, and halting desert dunes expansion in Rajasthan. **[1 Mark]**

• **Wildlife Conservation Network:** Establishing protected wildlife networks including National Parks, Wildlife Sanctuaries, Tiger Reserves (Project Tiger), and Biosphere Reserves under the Wildlife Protection Act 1972. **[1 Mark]**